

Data Engineering Roadmap — Beginner to Intermediate

A 12-week, topic-wise path from zero to job-ready. This is the order I'd learn things in if I were starting again — SQL first, then Python, then the modern data stack (Snowflake, dbt, Airflow), then the tooling around it (Git, AWS basics).

Two weeks per major topic is enough to go from "I don't know this" to "I can use this in a real pipeline." One week each for Git and AWS basics — you don't need depth there yet, just enough to not get stuck.

Week 1–2: SQL

Week 1 — Basics

- SELECT, WHERE, ORDER BY, LIMIT
- GROUP BY and HAVING
- Aggregate functions: COUNT, SUM, AVG, MIN, MAX
- Joins: INNER, LEFT, RIGHT, FULL OUTER
- Basic subqueries

Week 2 — Intermediate

- Window functions: ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), LEAD()
- Common Table Expressions (WITH clauses)
- CASE WHEN logic
- UNION / UNION ALL / INTERSECT

- Basic query optimization: reading an execution plan, why indexes matter

Practice: Solve 30–40 problems on a SQL practice site, mixing basics and window functions daily.

Week 3–4: Python

Week 3 — Basics

- Syntax, variables, data types
- Lists, dictionaries, sets, tuples
- Loops and conditionals
- Functions and scope
- Reading/writing files, working with CSV and JSON

Week 4 — Intermediate

- pandas basics: DataFrames, filtering, grouping, merging
- Virtual environments and pip
- Calling APIs with requests
- Error handling (try/except)
- Writing reusable functions and simple modules/scripts (not notebooks — actual .py files)

Practice: Take a messy CSV and clean it, transform it, and output a summary — end to end, in a script.

Week 5–6: Snowflake

Week 5 — Basics

- Architecture: storage and compute are separate
- Databases, schemas, tables
- Virtual warehouses — what they are, sizing
- Loading data: COPY INTO, internal vs external stages
- Basic role-based access (who can see/do what)

Week 6 — Intermediate

- Query performance: clustering keys, pruning
- Time Travel and zero-copy cloning
- Streams and Tasks (for change tracking and automation)
- Semi-structured data: VARIANT type, querying JSON
- Cost control: auto-suspend/resume, right-sizing warehouses

Practice: Load a raw dataset via COPY INTO, then write queries using Time Travel to see historical states.

Week 7–8: dbt

Week 7 — Basics

- Project structure: models/, sources.yml
- Materializations: view, table, incremental
- Basic Jinja templating ({{ }}, {% %})
- Generic tests: not_null, unique, relationships

Week 8 — Intermediate

- Incremental models and merge strategies
- Snapshots (for SCD Type 2 — tracking historical changes)
- Macros and reusable Jinja logic

- dbt packages (e.g. dbt_utils)
- Documentation and lineage graphs (dbt docs generate)

Practice: Build a small staging → intermediate → marts pipeline on sample data, with at least one incremental model and one snapshot.

Week 9–10: Airflow

Week 9 — Basics

- DAG concepts: nodes, dependencies, scheduling
- Common operators (PythonOperator, BashOperator)
- Task dependencies (>>, <<)
- Reading the Airflow UI (Graph view, logs)

Week 10 — Intermediate

- Sensors (waiting on external conditions)
- XComs (passing small data between tasks)
- Retries, alerting on failure
- Dynamic DAG generation
- Orchestrating dbt runs and Snowflake loads from Airflow
- Backfilling historical runs

Practice: Build a DAG that ingests a file, triggers a dbt run, and sends a notification on failure.

Week 11: Git & GitHub

- Commits, branches, merging
- Pull requests and code review basics
- Resolving merge conflicts

- .gitignore and why secrets should never be committed
 - GitHub Actions basics (a simple CI workflow that runs on push)
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Week 12: AWS Basics

- S3: buckets, storage classes, permissions
 - IAM: roles vs. users, least-privilege policies
 - Lambda: what serverless functions are, simple triggers
 - Snowpipe + S3 integration (event-driven ingestion)
 - CloudWatch: basic logging and monitoring
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After this roadmap

- AZ-900 (Azure Fundamentals) and AWS CLF-C02 (Cloud Practitioner) are good, low-effort certifications to validate cloud basics once you've covered Week 12.
- The Snowflake University free courses are worth doing alongside Weeks 5–6 — they map almost exactly to this order.
- Once you're through Week 12, the best next step isn't more theory — it's building one end-to-end project (ingestion → transformation → warehouse → a dashboard) and putting it on your GitHub.

Good luck — go one topic at a time, and don't skip the practice sections